STABILIZER-PRESERVING ETALE MAPS OF FINITE-FLAT GROUPOIDS DESCEND TO ETALE INVARIANT QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. Public299 treats inertia invariants of a finite abstract group acting on a ring. Here two finite-flat affine groupoids and a stabilizer-preserving morphism are given, and the conclusion includes a Cartesian atlas square.

Editorial note. Public1742 constructs an fppf invariant quotient for an equivalence relation; it does not establish this etale change theorem with stabilizers.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Affine groupoid schemes, finite locally free maps, strict henselization and etale descent, finite invariant-ring quotients and stabilizers. The source-local integrality, invariant base change and point-orbit arguments are retained.

Source reference. Original equation 03BF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4346–4346. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03BI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4349–4349. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03BH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4348–4348. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00DX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 514–514. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00GQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 616–616. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01S1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2322–2322. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

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concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00FL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 575–575. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00DS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 508–508. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01Z4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2579–2579. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 081E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10214–10214. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0476: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5497–5497. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00GA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 600–600. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09XK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12533–12533. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09XJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12532–12532. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09XI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12531–12531. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02GW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3243–3243. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3403–3403. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let S be a scheme. Let $f : (U', R', s', t') \to (U, R, s, t, c)$ be a morphism of groupoid schemes over S.*

(1) *U, R, U', R' are affine,*
(2) *s, t, s', t' are finite locally free,*
(3) *the diagrams*

$$\begin{array}{ccc} R' & \xrightarrow{f} & R \\ {\scriptstyle s'}\downarrow & & \downarrow{\scriptstyle s} \\ U' & \xrightarrow{f} & U \end{array} \qquad \begin{array}{ccc} R' & \xrightarrow{f} & R \\ {\scriptstyle t'}\downarrow & & \downarrow{\scriptstyle t} \\ U' & \xrightarrow{f} & U \end{array} \qquad \begin{array}{ccc} G' & \xrightarrow{f} & G \\ \downarrow & & \downarrow \\ U' & \xrightarrow{f} & U \end{array}$$

are cartesian where G and G' are the stabilizer group schemes, and

(4) *$f : U' \to U$ is étale.*

Then the map $C \to C'$ from the R-invariant functions on U to the R'-invariant functions on U' is étale and $U' = \mathrm{Spec}(C') \times_{\mathrm{Spec}(C)} U$.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(A)$ and $R = \mathrm{Spec}(B)$ are affine and $s, t : R \to U$ finite locally free. Let $C \subset A$ be as in (03BF). Then A is integral over C.*

Proof. First, by Lemma 03BI we know that (U, R, s, t, c) is a disjoint union of groupoid schemes (U_r, R_r, s, t, c) such that each $s, t : R_r \to U_r$ has constant rank r. As U is quasi-compact, we have $U_r = \emptyset$ for almost all r. It suffices to prove the lemma for each (U_r, R_r, s, t, c) and hence we may assume that s, t are finite locally free of rank r.

Assume that s, t are finite locally free of rank r. Let $f \in A$. Consider the element $x - f \in A[x]$, where we think of x as the coordinate on $\mathbf{A}^1$. Since

$$(U \times \mathbf{A}^1, R \times \mathbf{A}^1, s \times \mathrm{id}_{\mathbf{A}^1}, t \times \mathrm{id}_{\mathbf{A}^1}, c \times \mathrm{id}_{\mathbf{A}^1})$$

is also a groupoid scheme with finite source and target, we may apply Lemma 03BH to it and we see that $P(x) = \mathrm{Norm}_{s^\sharp}(t^\sharp(x - f))$ is an element of $C[x]$. Because $s^\sharp : A \to B$ is finite locally free of rank r we see that P is monic of degree r. Moreover $P(f) = 0$ by Cayley-Hamilton (Algebra, Lemma 00DX). More precisely, Cayley-Hamilton says the polynomial $s^\sharp(P) \in B[x]$ annihilates the element $t^\sharp(f)$. Since the coefficients of P are in C we get that $t^\sharp(P(f)) = 0$ in B. Since t is faithfully flat, we get $P(f) = 0$. □

Lemma 0.3. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(A)$ and $R = \mathrm{Spec}(B)$ are affine and $s, t : R \to U$ finite locally free. Let $C \subset A$ be as in (03BF). Let $C \to C'$ be a ring map, and set $U' = \mathrm{Spec}(A \otimes_C C')$, $R' = \mathrm{Spec}(B \otimes_C C')$. Then*

(1) *The maps s, t, c induce maps s', t', c' such that (U', R', s', t', c') is a groupoid scheme. Let $C^1 \subset A'$ be the R'-invariant functions on U'.*

(2) *The canonical map $\varphi : C' \to C^1$ satisfies*
   (a) *for every $f \in C^1$ there exists an $n > 0$ and a polynomial $P \in C'[x]$ whose image in $C^1[x]$ is $(x - f)^n$, and*
   (b) *for every $f \in \mathrm{Ker}(\varphi)$ there exists an $n > 0$ such that $f^n = 0$.*

(3) *If $C \to C'$ is flat then φ is an isomorphism.*

Proof. The proof of part (1) is omitted. Let us denote $A' = A \otimes_C C'$ and $B' = B \otimes_C C'$. Then we have

$$C^1 = \{a \in A' \mid (t')^\sharp(a) = (s')^\sharp(a)\} = \{a \in A \otimes_C C' \mid t^\sharp \otimes 1(a) = s^\sharp \otimes 1(a)\}.$$

In other words, C^1 is the kernel of the difference map $(t^\sharp - s^\sharp) \otimes 1$ which is just the base change of the C-linear map $t^\sharp - s^\sharp : A \to B$ by $C \to C'$. Hence (3) follows.

Proof of part (2)(b). Since $C \to A$ is integral (Lemma 0.2) and injective we see that $\mathrm{Spec}(A) \to \mathrm{Spec}(C)$ is surjective, see Algebra, Lemma 00GQ. Thus also $\mathrm{Spec}(A') \to \mathrm{Spec}(C')$ is surjective as a base change of a surjective morphism (Morphisms, Lemma 01S1). Hence $\mathrm{Spec}(C^1) \to \mathrm{Spec}(C')$ is surjective also. This implies (2)(b) holds for example by Algebra, Lemma 00FL.

Proof of part (2)(a). By Lemma 03BI our groupoid scheme (U, R, s, t, c) decomposes as a finite disjoint union of groupoid schemes (U_r, R_r, s, t, c) such that $s, t : R_r \to U_r$ are finite locally free of rank r. Pulling back by $U' = \mathrm{Spec}(C') \to U$ we obtain a similar decomposition of U' and $U^1 = \mathrm{Spec}(C^1)$. We will show in the next paragraph that (2)(a) holds for the corresponding system of rings $A_r, B_r, C_r, C'_r, C^1_r$ with $n = r$. Then given $f \in C^1$ let $P_r \in C_r[x]$ be the polynomial whose image in $C^1_r[x]$ is the image of $(x - f)^r$. Choosing a sufficiently divisible integer n we see that there is a polynomial $P \in C'[x]$ whose image in $C^1[x]$ is $(x - f)^n$; namely, we take P to be the unique element of $C'[x]$ whose image in $C'_r[x]$ is $P_r^{n/r}$.

In this paragraph we prove (2)(a) in case the ring maps $s^\sharp, t^\sharp : A \to B$ are finite locally free of a fixed rank r. Let $f \in C^1 \subset A' = A \otimes_C C'$. Choose a flat C-algebra D and a surjection $D \to C'$. Choose a lift $g \in A \otimes_C D$ of f. Consider the polynomial

$$P = \mathrm{Norm}_{s^\sharp \otimes 1}(t^\sharp \otimes 1(x - g))$$

in $(A \otimes_C D)[x]$. By Lemma 03BH and part (3) of the current lemma the coefficients of P are in D (compare with the proof of Lemma 0.2). On the other hand, the image of P in $(A \otimes_C C')[x]$ is $(x - f)^r$ because $t^\sharp \otimes 1(x - f) = s^\sharp(x - f)$ and $s^\sharp$ is finite locally free of rank r. This proves what we want with P as in the statement (2)(a) given by the image of our P under the map $D[x] \to C'[x]$. □

Lemma 0.4. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(A)$ and $R = \mathrm{Spec}(B)$ are affine and $s, t : R \to U$ finite locally free. Let $C \subset A$ be as in (03BF). Then $U \to M = \mathrm{Spec}(C)$ has the following properties:*

(1) *the map on points $|U| \to |M|$ is surjective and $u_0, u_1 \in |U|$ map to the same point if and only if there exists a $r \in |R|$ with $t(r) = u_0$ and $s(r) = u_1$, in a formula*
$$|M| = |U|/|R|$$

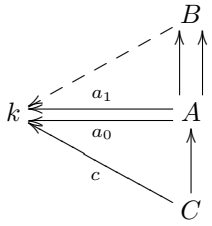
(2) *for any algebraically closed field k we have*
$$M(k) = U(k)/R(k)$$

Proof. Since $C \to A$ is integral (Lemma 0.2) and injective we see that $\mathrm{Spec}(A) \to \mathrm{Spec}(C)$ is surjective, see Algebra, Lemma 00GQ. Thus $|U| \to |M|$ is surjective.

Let k be an algebraically closed field and let $C \to k$ be a ring map. Since surjective morphisms are preserved under base change (Morphisms, Lemma 01S1) we see that

$A \otimes_C k$ is not zero. Now $k \subset A \otimes_C k$ is a nonzero integral extension. Hence any residue field of $A \otimes_C k$ is an algebraic extension of k, hence equal to k. Thus we see that $U(k) \to M(k)$ is surjective.

Let $a_0, a_1 : A \to k$ be two ring maps. If there exists a ring map $b : B \to k$ such that $a_0 = b \circ t^\sharp$ and $a_1 = b \circ s^\sharp$ then we see that $a_0|_C = a_1|_C$ by definition. Thus the map $U(k) \to M(k)$ equalizes the two maps $R(k) \to U(k)$. Conversely, suppose that $a_0|_C = a_1|_C$. Let us name this algebra map $c : C \to k$. Consider the diagram



If we can construct a dotted arrow making the diagram commute, then the proof of part (2) of the lemma is complete. Since $s : A \to B$ is finite there exist finitely many ring maps $b_1, \ldots, b_n : B \to k$ such that $b_i \circ s^\sharp = a_1$. If the dotted arrow does not exist, then we see that none of the $a_i' = b_i \circ t^\sharp$, $i = 1, \ldots, n$ is equal to a_0. Hence the maximal ideals

$$\mathfrak{m}_i' = \mathrm{Ker}(a_i' \otimes 1 : A \otimes_C k \to k)$$

of $A \otimes_C k$ are distinct from $\mathfrak{m} = \mathrm{Ker}(a_0 \otimes 1 : A \otimes_C k \to k)$. By Algebra, Lemma 00DS we would get an element $f \in A \otimes_C k$ with $f \in \mathfrak{m}$, but $f \notin \mathfrak{m}_i'$ for $i = 1, \ldots, n$. Consider the norm

$$g = \mathrm{Norm}_{s^\sharp \otimes 1}(t^\sharp \otimes 1(f)) \in A \otimes_C k$$

By Lemma 03BH this lies in the invariants $C^1 \subset A \otimes_C k$ of the base change groupoid (base change via the map $c : C \to k$). On the one hand, $a_1(g) \in k^*$ since the value of $t^\sharp(f)$ at all the points (which correspond to $b_1, \ldots, b_n$) lying over a_1 is invertible (insert future reference on property determinant here). On the other hand, since $f \in \mathfrak{m}$, we see that f is not a unit, hence $t^\sharp(f)$ is not a unit (as $t^\sharp \otimes 1$ is faithfully flat), hence its norm is not a unit (insert future reference on property determinant here). We conclude that C^1 contains an element which is not nilpotent and not a unit. We will now show that this leads to a contradiction. Namely, apply Lemma 0.3 to the map $c : C \to C' = k$, then we see that the map of k into the invariants C^1 is injective and moreover, that for any element $x \in C^1$ there exists an integer $n > 0$ such that $x^n \in k$. Hence every element of C^1 is either a unit or nilpotent.

We still have to finish the proof of (1). We already know that $|U| \to |M|$ is surjective. It is clear that $|U| \to |M|$ is $|R|$-invariant. Finally, suppose $u_0, u_1 \in U$ maps to the same point $m \in M$. Then the induced field extensions $\kappa(u_0)/\kappa(m)$ and $\kappa(u_1)/\kappa(m)$ are algebraic (as A is integral over C as used above). Hence if k is an algebraic closure of $\kappa(m)$ then we can find $\kappa(m)$-embeddings $\overline{u}_0 : \kappa(u_0) \to k$ and $\overline{u}_1 : \kappa(u_1) \to k$. These determine k-valued points $\overline{u}_0, \overline{u}_1 \in U(k)$ mapping to the same point of $M(k)$. By part (2) we see that there exists a point $\overline{r} \in R(k)$ with $s(\overline{r}) = \overline{u}_0$ and $t(\overline{r}) = \overline{u}_1$. The image $r \in R$ of $\overline{r}$ is a point with $s(r) = u_0$ and $t(r) = u_1$ as desired. □

Proof. Set $M = \mathrm{Spec}(C)$ and $M' = \mathrm{Spec}(C')$. Write $U = \mathrm{Spec}(A)$, $U' = \mathrm{Spec}(A')$, $R = \mathrm{Spec}(B)$, and $R' = \mathrm{Spec}(B')$. We will use the results of Lemmas 0.2, 0.3, and 0.4 without further mention.

Assume C is a strictly henselian local ring. Let $p \in M$ be the closed point and let $p' \in M'$ map to p. Claim: in this case there is a disjoint union decomposition $(U', R', s', t', c') = (U, R, s, t, c) \amalg (U'', R'', s'', t'', c'')$ over (U, R, s, t, c) such that for

the corresponding disjoint union decomposition $M' = M \amalg M''$ over M the point p' corresponds to $p \in M$.

The claim implies the lemma. Suppose that $M_1 \to M$ is a flat morphism of affine schemes. Then we can base change everything to M_1 without affecting the hypotheses (1) – (4). From Lemma 0.3 we see M_1, resp. M_1' is the spectrum of the R_1-invariant functions on U_1, resp. the R_1'-invariant functions on U_1'. Suppose that $p' \in M'$ maps to $p \in M$. Let M_1 be the spectrum of the strict henselization of $\mathcal{O}_{M,p}$ with closed point $p_1 \in M_1$. Choose a point $p_1' \in M_1'$ mapping to p_1 and p'. From the claim we get

$$(U_1', R_1', s_1', t_1', c_1') = (U_1, R_1, s_1, t_1, c_1) \amalg (U_1'', R_1'', s_1'', t_1'', c_1'')$$

and correspondingly $M_1' = M_1 \amalg M_1''$ as a scheme over M_1. Write $M_1 = \operatorname{Spec}(C_1)$ and write $C_1 = \operatorname{colim} C_i$ as a filtered colimit of étale C-algebras. Set $M_i = \operatorname{Spec}(C_i)$. The $M_1 = \lim M_i$ and similarly for the other schemes. By Limits, Lemmas 01Z4 and 081E we can find an i such that

$$(U_i', R_i', s_i', t_i', c_i') = (U_i, R_i, s_i, t_i, c_i) \amalg (U_i'', R_i'', s_i'', t_i'', c_i'')$$

We conclude that $M_i' = M_i \amalg M_i''$. In particular $M' \to M$ becomes étale at a point over p' after an étale base change. This implies that $M' \to M$ is étale at p' (for example by Morphisms, Lemma 0476). We will prove $U' \cong M' \times_M U$ after we prove the claim.

Proof of the claim. Observe that U_p and $U'_{p'}$ have finitely many points. For $u \in U_p$ we have $\kappa(u)/\kappa(p)$ is algebraic, hence $\kappa(u)$ is separably closed. As $U' \to U$ is étale, we conclude the morphism $U'_{p'} \to U_p$ induces isomorphisms on residue field extensions. Let $u' \in U'_{p'}$ with image $u \in U_p$. By assumption (3) the morphism of scheme theoretic fibres $(s')^{-1}(u') \to s^{-1}(u)$, $(t')^{-1}(u') \to t^{-1}(u)$, and $G'_{u'} \to G_u$ are isomorphisms. Observing that $U_p = t(s^{-1}(u))$ (set theoretically) we conclude that the points of $U'_{p'}$ surject onto the points of U_p. Suppose that u'_1 and u'_2 are points of $U'_{p'}$ mapping to the same point u of U_p. Then there exists a point $r' \in R'_{p'}$ with $s'(r') = u'_1$ and $t'(r') = u'_2$. Consider the two towers of fields

$$\kappa(r')/\kappa(u'_1)/\kappa(u)/\kappa(p) \quad \kappa(r')/\kappa(u'_2)/\kappa(u)/\kappa(p)$$

whose "ends" are the same as the two "ends" of the two towers

$$\kappa(r')/\kappa(u'_1)/\kappa(p')/\kappa(p) \quad \kappa(r')/\kappa(u'_2)/\kappa(p')/\kappa(p)$$

These two induce the same maps $\kappa(p') \to \kappa(r')$ as $(U'_{p'}, R'_{p'}, s', t', c')$ is a groupoid over p'. Since $\kappa(u)/\kappa(p)$ is purely inseparable, we conclude that the two induced maps $\kappa(u) \to \kappa(r')$ are the same. Therefore r' maps to a point of the fibre G_u. By assumption (3) we conclude that $r' \in (G')_{u'_1}$. Namely, we may think of G as a closed subscheme of R viewed as a scheme over U via s and use that the base change to U' gives $G' \subset R'$. In particular we have $u'_1 = u'_2$. We conclude that $U'_{p'} \to U_p$ is a bijective map on points inducing isomorphisms on residue fields. It follows that $U'_{p'}$ is a finite set of closed points (Algebra, Lemma 00GA) and hence $U'_{p'}$ is closed in U'. Let $J' \subset A'$ be the radical ideal cutting out $U'_{p'}$ set theoretically.

Second part proof of the claim. Let $\mathfrak{m} \subset C$ be the maximal ideal. Observe that $(A, \mathfrak{m}A)$ is a henselian pair by More on Algebra, Lemma 09XK. Let $J = \sqrt{\mathfrak{m}A}$. Then (A, J) is a henselian pair (More on Algebra, Lemma 09XJ) and the étale ring map $A \to A'$ induces an isomorphism $A/J \to A'/J'$ by our deliberations above. We conclude that $A' = A \times A''$ by More on Algebra, Lemma 09XI. Consider the corresponding disjoint union decomposition $U' = U \amalg U''$. The open $(s')^{-1}(U)$ is the set of points of R' specializing to a point of $R'_{p'}$. Similarly for $(t')^{-1}(U)$.

Similarly we have $(s')^{-1}(U'') = (t')^{-1}(U'')$ as this is the set of points which do not specialize to $R'_{p'}$. Hence we obtain a disjoint union decomposition

$$(U', R', s', t', c') = (U, R, s, t, c) \amalg (U'', R'', s'', t'', c'')$$

This immediately gives $M' = M \amalg M''$ and the proof of the claim is complete.

We still have to prove that the canonical map $U' \to M' \times_M U$ is an isomorphism. It is an étale morphism (Morphisms, Lemma 02GW). On the other hand, by base changing to strictly henselian local rings (as in the third paragraph of the proof) and using the bijectivity $U'_{p'} \to U_p$ established in the course of the proof of the claim, we see that $U' \to M' \times_M U$ is universally bijective (some details omitted). However, a universally bijective étale morphism is an isomorphism (Descent, Lemma 02LC) and the proof is complete. □